MATHEMATICS

GENERAL NOTATION, SETS AND PROBABILITY

General notation

Symbol	Meaning	Examples
$=$	Equal	$2+3=5$ $\quad 4-1=3$
$\neq$	Not equal	$2\neq 3$ $\quad 5+1\neq 7$
$\approx$	Approximately	$\pi\approx 3.14$ $\quad \sqrt{2}\approx 1.41$
$<$ $>$	Less than Greater than	$2<3$ $\quad 5>4$
$\not<$ $\not>$	Not Less than Not Greater than	$2\not<2$ $\quad 4\not>5$
$\leq$ $\geq$	Less/Greater than or equal to	$2\leq 2$ $\quad 5\geq 4$
$\not\leq$ $\not\geq$	Not Less/Greater than or equal to	$3\not\leq 2$ $\quad 4\not\geq 5$
$\pm$	Plus or minus: *two choices*	$3\pm 1=4$ or 2 $\quad -2\pm 3=1$ or -5
$ab=a(b)=a\cdot b=a\times b$	Multiplication: *juxtaposition, parentheses, dot, or cross.*	$3x=3\cdot x=3\times x$ $\quad 2(4)=2\cdot 4=2\times 4$
$\frac{a}{b}=a/b=a\div b$	Division: *fraction, slash, or division sign* $(b\neq 0)$.	$\frac{12}{3}=12/3=12\div 3=4$ $\quad \frac{x}{5}=x/5=x\div 5=\frac{1}{5}x=0.2x$
$\lvert a\rvert$	Absolute value: *distance from* a *to* 0	$\lvert -5\rvert=5$ $\quad \lvert 3\rvert=3$ $\quad \lvert 0\rvert=0$
a^2	Squared; *multiply* a *by itself.*	$3^2=3\cdot 3=9$ $\quad (-2)^2=(-2)(-2)=4$
a^3	Cubed; *multiply three factors of* a.	$2^3=2\cdot 2\cdot 2=8$ $\quad (-2)^3=(-2)(-2)(-2)=-8$
$\sqrt{a}=a^{1/2}$	Square root $\quad a\geq 0$	$\sqrt{9}=3$ (not ± 3)
$\sqrt[3]{a}=a^{1/3}$	Cube root $\quad a\in\mathbb{R}$	$\sqrt[3]{-8}=-2$
a^n	Exponentiation: n *factors of* a	$2^4=2\cdot 2\cdot 2\cdot 2=16$ $\quad (-2)^5=-32$
$\sqrt[n]{a}=a^{1/n}$	nth root.	$\sqrt[4]{16}=16^{\frac{1}{4}}=2$ $\quad \sqrt[5]{-32}=-2=(-32)^{\frac{1}{5}}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set *Ignores order and repetition.*	$\{2,4\}=\{4,2\}$ $\{2,2,4\}=\{2,4\}$
$\in$	Element of	$2\in\{1,2\}$
$\notin$	Not an element of	$3\notin\{1,2\}$
$\{x:p(x)\}$	Set-builder notation $x\in A$ *such that* $p(x)$ *holds.*	$\{x\in\mathbb{R}:x>0\}$ Positive real numbers
$\varnothing$	Empty set *No elements.*	$\lvert\varnothing\rvert=0$
$\lvert A\rvert$	Cardinality *Number of elements in finite* A.	$\lvert\{2,4,6\}\rvert=3$
$A\subseteq B$	Subset *All of* A *in* B*; may equal* B.	$\{1,2\}\subseteq\{1,2\}$
$A\subsetneq B$	Proper subset $A\subseteq B$, $A\neq B$.	$\{1\}\subsetneq\{1,2\}$
$A\cup B$	Union *In either or both.*	$\{1\}\cup\{2\}=\{1,2\}$
$A\cap B$	Intersection *In both.*	$\{1,2\}\cap\{2\}=\{2\}$
$A\setminus B$	Difference *In* A*, outside* B.	$\{1,2\}\setminus\{2\}=\{1\}$
$\mathbb{N}$	Naturals	$\mathbb{N}=\{1,2,3,...\}$
$\mathbb{Z}$	Integers	$Z=\{...,-2,-1,0,1,2,...\}$
$\mathbb{Q}$	Rationals p/q; $p,q\in\mathbb{Z}$, $q\neq 0$.	$\frac{1}{2},-3\in\mathbb{Q}$
$\mathbb{R}$	Reals *The number line.*	$\sqrt{2},\pi,\frac{1}{2},-3\in\mathbb{R}$
$\mathbb{R}\setminus\mathbb{Q}$	Irrationals	$\sqrt{2},\pi\in\mathbb{R}\setminus\mathbb{Q}$
(a,b)	Open interval *Excludes endpoints.*	$x\in(1,3):1<x<3$
$[a,b]$	Closed interval *Includes endpoints.*	$x\in[1,3]:1\leq x\leq 3$
$[a,b)$	Half-open interval *Includes left endpoint only.*	$2\in[2,5),5\notin[2,5)$
$(a,b]$	Half-open interval *Includes right endpoint only.*	$2\notin(2,5],5\in(2,5]$
∞	Positive infinity *Unbounded above.*	$[0,\infty)$
$-\infty$	Negative infinity *Unbounded below.*	$(-\infty,0]$

Intervals assume $a<b$. Infinity is not a real number; use parentheses there. Some texts include 0 in $\mathbb{N}$; check the convention. Dots: see Poster 2. Bars: absolute value for numbers, cardinality for finite sets.

Probability and events

Example contexts
Fair die: $\Omega=\{1,2,3,4,5,6\}$.
$A=\{1,3,5\}$ (odd), $B=\{1,2\}$, $C=\{2,4,6\}$.
Die examples below use these events.
Independent fair coin tosses: E: first is heads; F: second is heads.

Symbol	Meaning	Examples
Ω	Sample space *All possible outcomes.*	$\Omega=\{1,2,3,4,5,6\}$
ω	Outcome *One possible result.*	$\omega=3\in\Omega$
$A\subseteq\Omega$	Event *A set of outcomes.*	$A=\{1,3,5\}$
$P(A)$	Probability of A *Between 0 and 1.*	$0\leq P(A)\leq 1$ $P(\Omega)=1$ $P(\varnothing)=0$
$\frac{\lvert A\rvert}{\lvert\Omega\rvert}$	Classical probability *Equals* $P(A)$ *for finite, equally likely outcomes only.*	$P(A)=\frac{3}{6}=\frac{1}{2}$
A^c	Complement *Outcomes in* Ω *outside* A.	$A^c=\{2,4,6\}$ $P(A^c)=1-P(A)=\frac{1}{2}$
$A\cup B$	Union of events *Event* A *or* B*, or both.*	$A\cup B=\{1,2,3,5\}$
$A\cap B$	Intersection of events *Events* A *and* B *together.*	$A\cap B=\{1\}$
$A\cap C=\varnothing$	Mutually exclusive *Cannot occur together.*	$A\cap C=\varnothing$
$P(A\mid B)$	Conditional probability *A given* B*;* $P(B)>0$.	$P(A\mid B)=\frac{P(A\cap B)}{P(B)}=\frac{1/6}{2/6}=\frac{1}{2}$
$P(E\cap F)$	Independent events *Multiply their probabilities.*	$P(E\cap F)=P(E)P(F)=\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{4}$
$P(A\mid B)$	Independence *Iff this equals* $P(A)$*;* $P(B)>0$.	$P(E\mid F)=P(E)=\frac{1}{2}$

Keep the ideas separate.
Mutually exclusive events of positive probability are not independent: one occurring rules out the other.

The bar in $P(A\mid B)$ means "given"; in set-builder notation it means "such that". Context determines the reading.

Symbols acquire meaning from their mathematical context.

MATHEMATICS

FUNCTIONS, VARIABLES, SEQUENCES AND SUMS

Function notation

Example contexts
Example: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
Set and interval notation: see Poster 1.

Symbol	Meaning	Examples
f	Function *One output per domain input.*	$f(x) = x^2$
$f(x)$	Function value *Value at x; parentheses mark input, not multiplication.*	$f(3) = 3^2 = 9$
$f : A \to B$	Function mapping *Domain A to codomain B.*	$f : \mathbb{R} \to \mathbb{R}$
A	Domain *Allowed inputs.*	$A = \mathbb{R}$
B	Codomain *Declared output target.*	$B = \mathbb{R}$
$x \mapsto f(x)$	Mapping rule *Maps values; $\to$ links sets.*	$x \mapsto x^2$ $3 \mapsto 9$
$\mathrm{dom}(f)$	Domain *Allowed inputs.*	$\mathrm{dom}(f) = \mathbb{R}$
$\mathrm{range}(f)$	Range *Attained outputs in codomain.*	$\mathrm{range}(f) = [0, \infty)$ No negative outputs.

Letters and Greek symbols

Symbol	Name / description
$a, b, c;\ x, y, z$	Common lowercase Latin letters
A, B, C	Common uppercase Latin letters
α	Alpha
β	Beta
γ	Gamma
δ	Delta
Δ	Uppercase delta
θ	Theta
Θ	Uppercase theta
λ	Lambda
Λ	Uppercase lambda
μ	Mu
σ	Sigma
Σ	Uppercase sigma
ω	Omega
Ω	Uppercase omega *(Poster 1)*
π	Pi *Here, the circle constant $\pi \approx 3.14$.*

Names do not fix meanings. Σ is a letter; $\sum$ is the related summation operator (right), not an ordinary variable.

Variables

Letter roles
Roles depend on context: uppercase often names sets; Greek often names angles or numbers. Conventions are not permanent types.

Symbol	Meaning	Examples
x	Variable *May vary in context.*	$x \in \{-2, 0, 3.5\}$
m, b	Constants *Fixed in context.*	$y = mx + b$
x	Input variable *Input for a line.*	$y = mx + b$
y	Output variable *Output for that line.*	m, b fixed
n	Count *May count objects.*	$n \in \mathbb{N}$
θ	Angle *May name an angle in degrees ($^\circ$).*	$\theta = 30^\circ$
A	Set name *May name a set.*	$A = \{1, 2\}$
λ	Number *May name a number.*	$\lambda = 2$

A letter may be variable or constant. Values -2, 0, 3.5 are separate choices for x, not simultaneous equalities.

Lists, sequences and subscripts

Reading guide
A **sequence** is an ordered list of terms; a **series** is a sum of sequence terms. Here, parentheses show sequences; braces show sets.
Sequence examples: $a_i = 2i$, starting at $i = 1$.

Symbol	Meaning	Examples
$\ldots$	Ellipsis *Omitted list entries.*	$1, 2, 3, \ldots$
$\cdots$	Centered ellipsis *Omitted terms in a sum.*	$a_1 + a_2 + \cdots + a_n$
$1, 2, \ldots, n$	Finite list *Ends at n.*	$1, 2, 3, \ldots, 5$ $(n = 5)$
$1, 2, 3, \ldots$	Infinite list *No final term.*	$1, 2, 3, 4, 5, \ldots$
a_i	Indexed term *Term at position or label i.*	$a_3 = 2(3) = 6$
a_1	Indexed term *Term at index 1.*	$a_1 = 2$
a_n	Indexed term *Term at index n.*	$a_n = 2n$
$(a_i)_{i=1}^{4}$	Finite sequence *Four terms.*	$(a_i)_{i=1}^{4} = (2, 4, 6, 8)$
$(a_i)_{i=1}^{\infty}$	Infinite sequence *No last term.*	$(a_i)_{i=1}^{\infty} = (2, 4, 6, \ldots)$
$(2, 4)$	Sequence order *Order matters.*	$(2, 4) \neq (4, 2)$
$(2, 2, 4)$	Sequence repetition *Repetition matters.*	$(2, 2, 4) \neq (2, 4)$
A_1, A_2	Subscript labels *Subscripts may label cases.*	Alternatives A_1 and A_2.

Dots follow an established pattern; alone they do not define a unique rule. The finite list $1, 2, 3, \ldots, n$ assumes integer $n \geq 3$. An index is not multiplication or an exponent. Indices may start at 1, 0, or another integer: state the start. Sets ignore order and repetition (Poster 1).

Summation notation

Index convention
Indices run through consecutive integers between the bounds, **inclusive**. Here $n \in \mathbb{N}$, $n \geq 1$, and $a_i = 2i$ in the sequence examples.

Symbol	Meaning	Examples
$\sum_{i=1}^{n} a_i$	Summation *Sum of the indicated terms.*	$a_1 + a_2 + \cdots + a_n$
i	Index *Local to sum.*	$i = 1, 2, \ldots, n$
a_i	Summand *Expression summed.*	$a_i = 2i$
1	Lower bound *First index.*	Start at $i = 1$.
n	Upper bound *Last index.*	End at $i = n$.
a_n	Indexed term *Single term at n.*	$a_4 = 8$
S_n	Partial sum *Sum of first n terms; start at 1.*	$S_n = \sum_{i=1}^{n} a_i$
S_4	Fourth partial sum *Four terms for $a_i = 2i$.*	$S_4 = 2 + 4 + 6 + 8 = 20$
$\sum_{i=2}^{4} 3$	Constant sum *Constant; bounds 2 to 4.*	$\sum_{i=2}^{4} 3 = 3 + 3 + 3 = 9$
$\sum_{k=1}^{3} k^2$	Sum of squares *Squares; bounds 1 to 3.*	$\sum_{k=1}^{3} k^2 = 1 + 4 + 9 = 14$
$i,\ k$	Index renaming *Index names may change consistently.*	$\sum_{i=1}^{4} i = \sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10$

Rename the local index in both bounds and summand. Centered dots use the same omission convention as list dots.

Symbols acquire meaning from their mathematical context.